

MemoryPalace



MOMENTS WORTH KEEPING

Your brain already flags the moments that mattered.
We wired that flag to a camera.

HACKMIT 2026

HEALTHCARE TRACK

THE PROBLEM

You cannot know which ten seconds will matter.

RECORD IT ALL?

30 to 45 min

Measured battery life of continuous wearable video.

KEEP IT ALL?

0 hours

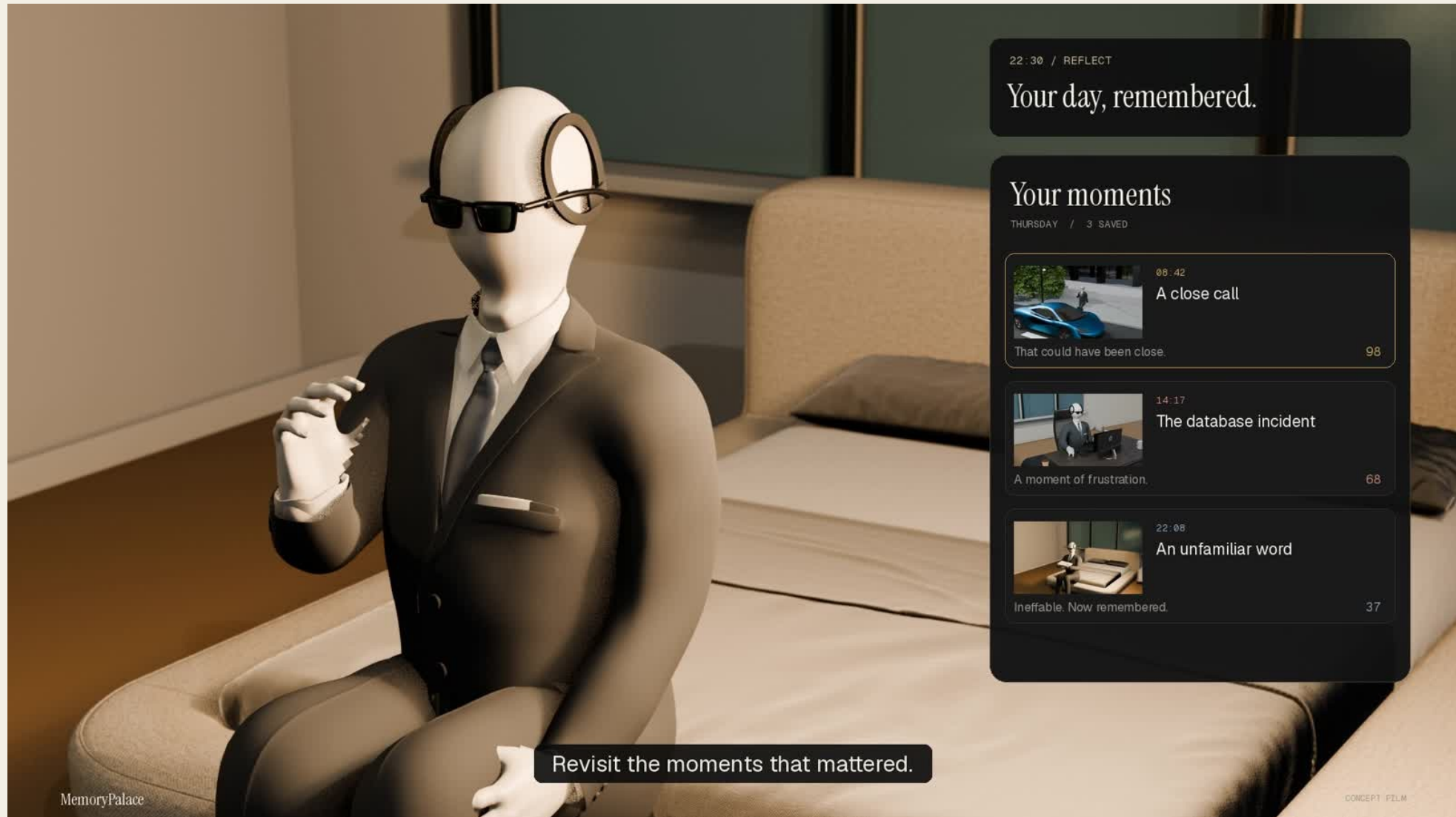
Nobody ever scrubs ten hours of their own footage.

WHAT WORKED

1 still/30 s

SenseCam helped dementia patients. No video at all.

Capture is solved. Deciding what is worth capturing is the product.



MEMORYPALACE

36 SECONDS

PLAY WITH SOUND

HOW A MOMENT IS CHOSEN

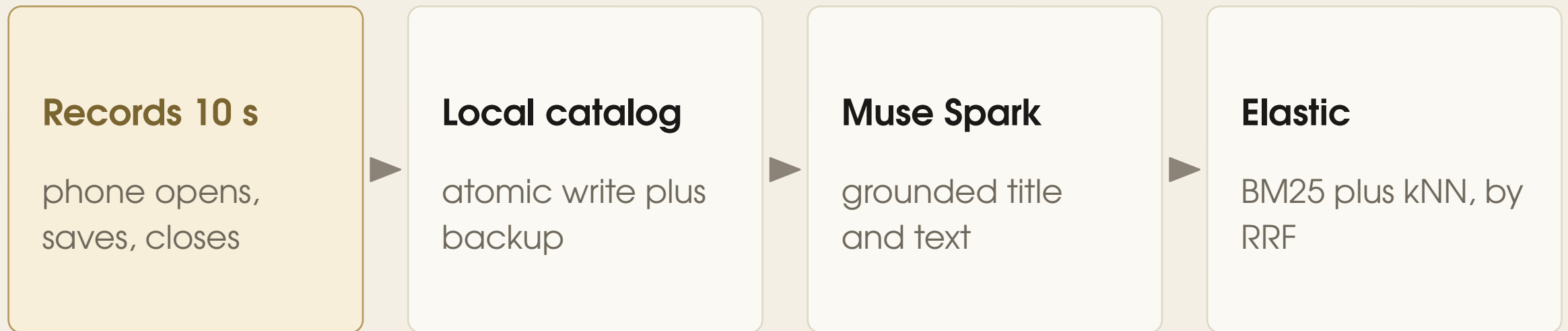
**The camera stays off until
your own baseline breaks.**



Only when the fourth window comes back elevated does the phone wake up.

WHAT HAPPENS TO THE CLIP

Committed locally first.
Everything after is enrichment.



A sponsor service outage degrades search. It can never lose a memory.

Why the baseline has to be yours

CALIBRATION

The threshold is universal. The baseline is not.



$$z = \frac{x - \mu}{\sigma}$$

Four in a row, and why they must not overlap

THE PERSISTENCE FILTER

“Four in a row” only counts if the four are independent.

hop 1 s

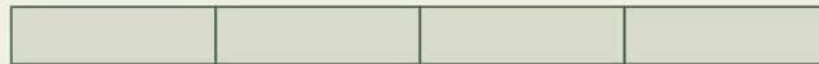


75 % shared data: two real looks

30.3

false alarms / hour

hop 4 s



no shared data: four real looks

1.3

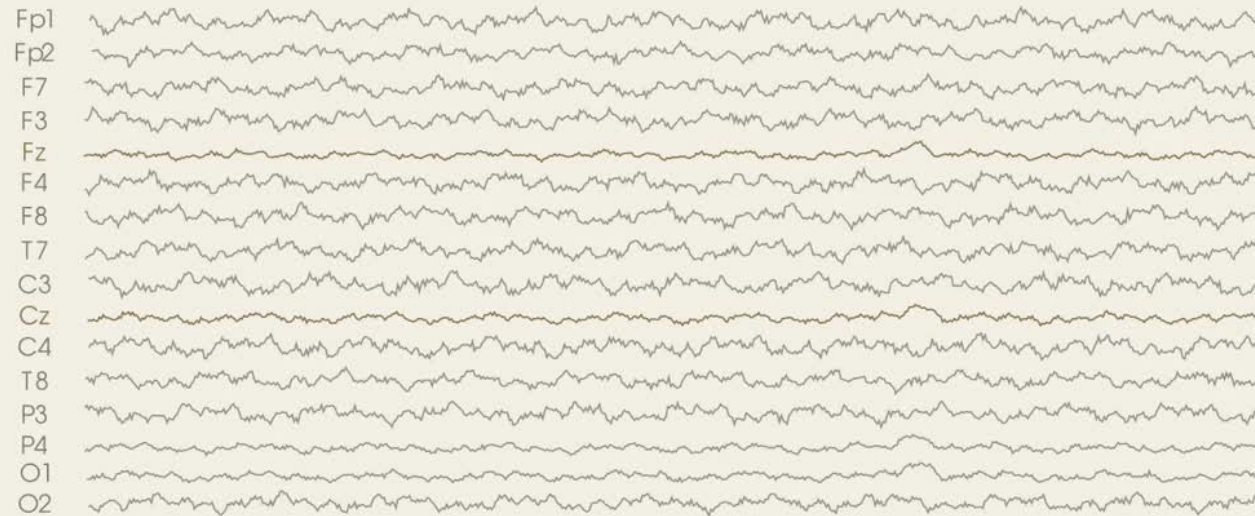
false alarms / hour

Same code. Same threshold. 24x fewer false alarms.

What is frozen before any model is fit

OPENNEURO DS006394 · CC0

Sixteen channels, 125 times a second.



33 participants · 56 usable recordings · left-earlobe reference

SPONSOR TRACK

Elastic

The retrieval layer: two kinds of remembering, fused.

WHAT WE BUILT

- Every moment indexed as one document on Elastic Cloud.
- Jina v5 Omni puts clips and queries in one vector space.
- BM25 and dense kNN merged by reciprocal rank fusion.
- EEG intensity and date bounds applied in both branches.

WHAT WE HAVE NOT BUILT

- Workflows and Agent Builder are future work, not done.
- No relevance tuning. RRF ranking is out of the box.
- Without credentials it falls back to local text search.

SPONSOR TRACK

Meta

Understanding and the agent: a clip becomes language.

WHAT WE BUILT

- Muse Spark 1.3 writes a grounded title per clip.
- Clips over 6 MB use the poster frame, and we log which.
- A web grounding pass resolves names, never the clip.
- Muse Voice Transcribe turns speech into the query.

WHAT WE HAVE NOT BUILT

- No Meta text to speech. Output is the browser voice.
- No physical glasses. The phone camera stands in.
- No emotion inference. Scene metadata only.

SPONSOR TRACK

VoloRidge

Public data, frozen before the test. Failures included.

WHAT WE BUILT

- Public CC0 EEG only. Every skipped recording is logged.
- Preprocessing and features locked before any model fit.
- Leave one out. The held out person picks no threshold.
- Three negative controls and an electrode ablation.

WHAT WE HAVE NOT BUILT

- No submission framing yet. The evidence exists.
- No live human validation. Synthetic EEG throughout.
- Our first continuous detector failed. We published it.

WHAT WE DO NOT CLAIM

Said first, before anyone asks.

NO HUMAN DETECTION

Every end to end run used synthetic EEG.

NOT A PROBABILITY

A workload proxy, not a likelihood.

NOT A DETECTION

We rank moments. We do not threshold.

NO EMOTION

An EEG spike does not name a feeling.

NO GLASSES

A phone camera stands in, marked as a demo.

WHAT IS NEXT

EEG is non stationary. So the model has to keep learning.

Spatial filters are tied to electrode positions, so weights never transfer between people. Retraining on the wearer is mandatory, which is why per user adaptation is the product and not a setup step. Every moment you confirm or dismiss is a label.

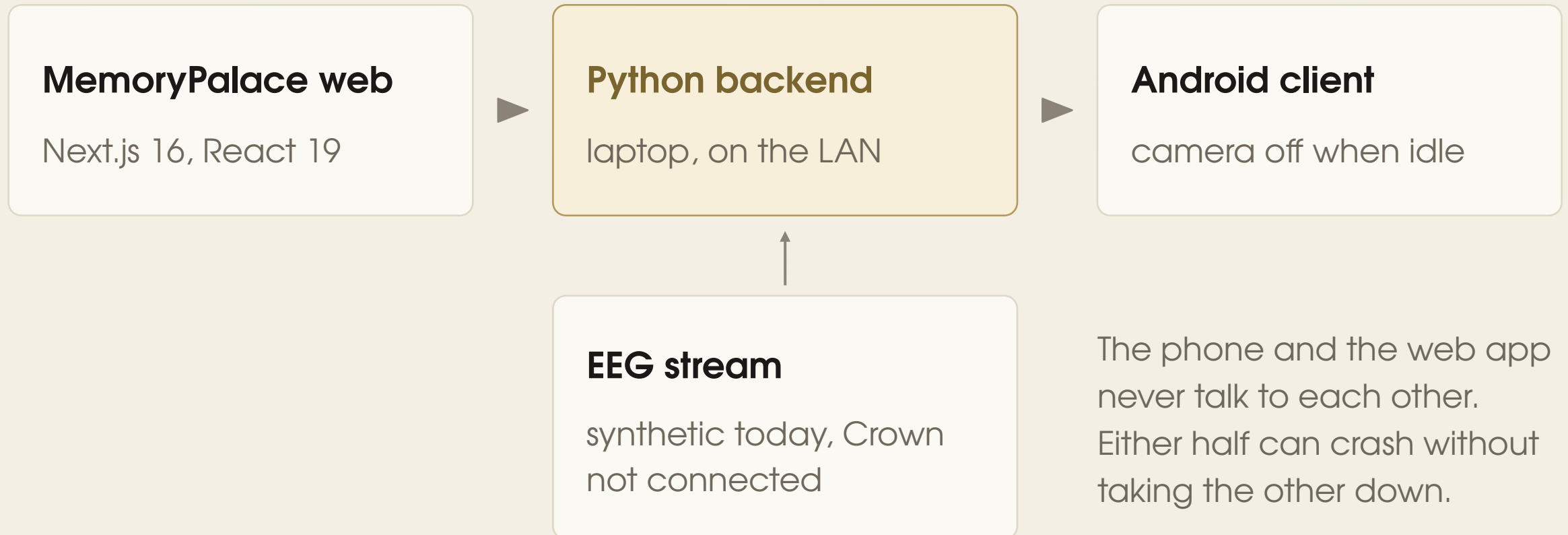
Connect the
Crown

Rolling pre spike
buffer

MongoDB
metadata layer

Learning curve vs
data

Three processes, and no direct phone to browser link.



The order, and what to say when it breaks.

THE ORDER

1. Pair the phone.
2. Open /live, simulate a spike.
3. Camera records ten seconds.
4. The moment appears in the gallery.
5. Search /explore for a phrase.
6. Click the dot to replay the clip.

IF IT BREAKS

- Nothing responds: you are on next dev. Use the build.
- Gallery empty: honest, nothing is seeded. Capture one.
- Elastic down: search falls back to local text, and says so.
- No wifi: run the backend in demo mode, no hardware.